

Fall 2024

Homework 4

STAT 656: Bayesian Data Analysis

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Gibbs Sampling for Probit Model

Consider N pairs of observations (x_i, y_i) , where $x_i \in \mathbf{R}^d$ and $y_i \in \{0, 1\}$. These are generated as follows:

$$z_i \sim \mathcal{N}(x_i\beta, 1) \quad y_i = \mathbb{1}(z_i > 0), \quad i = 1, \dots, N$$

This is thus the probit model from the Lecture slides. Write X , Y , and Z for the collection of x_i , y_i , and z_i 's. We will place a $\mathcal{N}(0, \Sigma_b)$ prior on $\beta \in \mathbf{R}^d$, and draw samples from the posterior $p(\beta, Z \mid X, Y)$. Observe that as in regression settings, we are really only interested in β , the Z 's only auxiliary variables to allow us to implement a Gibbs sampler. In other words, this is an example of a *data-augmentation algorithm*.

1. (7 points) Derive and write down the conditional distribution $p(Z \mid X, Y, \beta)$.
2. (7 points) Derive and write down the conditional distribution $p(\beta \mid X, Y, Z)$?
3. (1 points) Describe the overall Gibbs sampling algorithm, and how you would calculate the posterior mean and covariance of β .
4. (20 points) Write an R function that implement the VB algorithm for the probit model given in the slides. This should accept as input a dataset (X, Y) and return the approximations to the posterior distribution over β and the z 's. Recall how the algorithm proceeds: start with some arbitrary initial distribution $q(\beta)$ over β , use that to calculate the $q(z_i)$'s, use those to calculate $q(\beta)$, and repeat till these distributions stop changing.

Solution:

1. The conditional distribution $Z \mid X, Y, \beta$ is derived as follows:

$$\begin{aligned} \log p(Z \mid X, Y, \beta) &= \log p(\beta) + \sum_{i=1}^d \log p(y_i \mid z_i) + \log p(z_i \mid x_i) + C_1 \\ &= -\frac{1}{2}\beta^T \Sigma_b^{-1} \beta + \sum_{i=1}^d y_i \log \mathbb{1}(z_i > 0) + (1 - y_i) \log \mathbb{1}(z_i \leq 0) - \frac{1}{2}(z_i - x_i\beta)^2 + C_2 \end{aligned} \tag{1}$$

Therefore, the conditional distribution is given by:

$$p(Z \mid X, Y, \beta) \propto \begin{cases} \mathbb{1}(z_i > 0) \mathcal{N}(x_i\beta, 1) & \text{if } y_i = 1, \\ \mathbb{1}(z_i \leq 0) \mathcal{N}(x_i\beta, 1) & \text{otherwise.} \end{cases} \tag{2}$$

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2. Based on the results from equation (1), the conditional distribution of $\beta \mid X, Y, Z$ is derived by:

$$\begin{aligned}
\log p(\beta \mid X, Y, Z) &= -\frac{1}{2} \sum_{i=1}^d z_i^2 - 2z_i(x_i\beta) + x_i\beta\beta^\top x_i^\top + \beta^\top \Sigma_b^{-1} \beta + C_3 \\
&= -\frac{1}{2} \beta^\top \left(\sum_{i=1}^d x_i^\top x_i + \Sigma_b^{-1} \right) \beta - 2 \sum_{i=1}^d z_i x_i \beta + C_4 \\
&= -\frac{1}{2} \left(\beta - \Sigma_{\beta \mid X, Y, Z} \sum_{i=1}^d z_i x_i^\top \right)^\top \Sigma_{\beta \mid X, Y, Z}^{-1} \left(\beta - \Sigma_{\beta \mid X, Y, Z} \sum_{i=1}^d z_i x_i^\top \right) + C_5, \\
\text{where } \Sigma_{\beta \mid X, Y, Z} &= \left(\sum_{i=1}^d x_i^\top x_i + \Sigma_b^{-1} \right)^{-1}
\end{aligned}$$

Therefore, the conditional distribution is a multivariate normal distribution

$$p(\beta \mid X, Y, Z) \sim \mathcal{N} \left(\left(\sum_{i=1}^d x_i^\top x_i + \Sigma_b^{-1} \right)^{-1} \sum_{i=1}^d z_i x_i^\top, \left(\sum_{i=1}^d x_i^\top x_i + \Sigma_b^{-1} \right)^{-1} \right) \quad (2)$$

3. The Gibbs sampling algorithm is as follows:

- Initialize β and Z .
- Repeat the following steps until convergence or a maximum number of iterations:
 - Sample $Z \mid X, Y, \beta$ from the conditional distribution $p(Z \mid X, Y, \beta)$.
 - Sample $\beta \mid X, Y, Z$ from the conditional distribution $p(\beta \mid X, Y, Z)$. Therefore, with every sampled collection of Z , we can calculate the posterior mean and covariance of β according to the results equation (2):
- Calculate the covariance by $\left(\sum_{i=1}^d x_i^\top x_i + \Sigma_b^{-1} \right)^{-1}$.
- Calculate the mean by $\left(\sum_{i=1}^d x_i^\top x_i + \Sigma_b^{-1} \right)^{-1} \sum_{i=1}^d z_i x_i^\top$.